

GURUHLI MASHG'ULOT UCHUN O'QUV MATERIALLARI

2-Mavzu: PyQt5 paketi va QtDesigner dasturi yordamida GUI dasturlarini yaratish.

3-mashg'ulot. PyQt5 kutubxonasi. QLineEdit vidjeti.

O'quv savollari:

1. QLineEdit vidjeti;
2. setStyleSheet() metodi.

1. QLineEdit vidjeti

QLineEdit - bu matnning bir qatorini kiritish va tahrirlash imkonini beruvchi vidjet. Ushbu vidjetda Cencel va Repeat, Cut va Past hamda "Sudrab olib tashlash" funksiyalari mavjud.

```
import sys
from PyQt5.QtWidgets import (QWidget, QLabel,
                              QLineEdit, QApplication)

class Examples(QWidget):
    def __init__(self):
        super().__init__()

        self.initUI()

    def initUI(self):
        self.lbl = QLabel(self)
        qlineE = QLineEdit(self)

        qlineE.move(100, 100)
        self.lbl.move(100, 40)

        qlineE.textChanged[str].connect(self.changed)
        self.setGeometry(300, 300, 280, 170)
        self.setWindowTitle('QlineEdit Example')
        self.show()

    def changed(self, text):
        self.lbl.setText(Label text)
        self.lbl.adjustSize()

if __name__ == '__main__':

    app = QApplication(sys.argv)
    ex = Examples()
    sys.exit(app.exec_())
```

Ushbu misolda QLineEdit vidjeti va QLabel ko'rsatilgan. Tahrirlash paneliga kiritgan matnimiz darhol teg vidjetida ko'rsatiladi.

```
qlineE = QLineEdit(self)
```

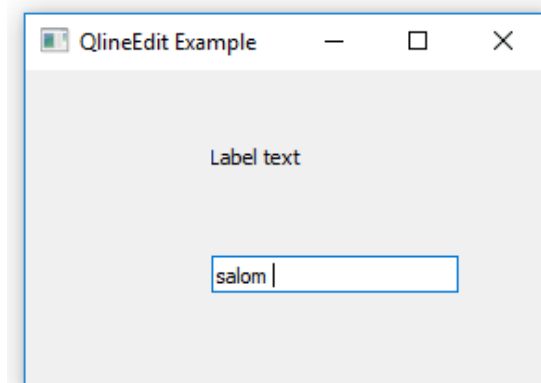
QLineEdit vidjetining `qline` obyekti yaratildi.

```
qlineE.textChanged[str].connect(self.changed)
```

Agar QLineEdit vidjetidagi matn o'zgarsa, `changed()` funksiyasi ishga tushadi.

```
def changed(self, text):  
    self.lbl.setText(text Label)  
    self.lbl.adjustSize()
```

`changed` funksiyasining ichida kiritilgan matnni teg vidjetiga o'rnatish buyrug'i joylashtirilgan. Yorliq o'lchamini matn uzunligiga mos ravishda o'zgartirish uchun `adjustSize()` metodidan foydalanilgan .



Yuqoridagi dastur ko'rinishi

PyQt QLineEdit

PyQt QLineEdit bir qatorli matnli vidjet yaratish imkonini beradi. Odatda, siz QLineEditma'lumotlarni kiritish shaklida foydalanasiz .

Amalda siz ko'pincha QLineEditvidjetni vidjet bilan ishlatasiz QLabel.

Vidjet yaratish uchun QLineEditquyidagi amallarni bajaring.

Birinchidan, moduldan import QLineEditqiling PyQt6.QtWidgets:

from PyQt6.QtWidgets import QLineEditKod tili: Python (python)

Ikkinchidan, QLineEditfoydalanadigan yangi obyekt yarating:

- Hech qanday dalil yo'q.
- Faqat ota-ona vidjeti bilan.
- Yoki birinchi argument sifatida standart satr qiymati bilan.

Masalan:

line_edit = QLineEdit('Default Value', self)Kod tili: Python (python)

Bundan tashqari, siz quyidagi qo'shimcha xususiyatlardan foydalanishingiz mumkin:

Metod	Turi	Ta'rifi
matn	ip	Satrni tahrirlash mazmuni
readOnly	Mantiqiy	To'g'ri yoki noto'g'ri. Agar rost bo'lsa, qatorni tahrir qilib bo'lmaydi
clearButtonEnabled	Mantiqiy	Aniq tugmani qo'shish to'g'ri

Metod	Turi	Ta'rifi
placeholderText	ip	Satni tahrirlash bo'sh bo'lganda paydo bo'ladigan matn
maxLength	butun son	Maksimal kiritilishi mumkin bo'lgan belgilar sonini belgilang
echoMode	QLineEdit.EchoMode	Matnni ko'rsatish usulini o'zgartiring, masalan, parol

PyQt QLineEdit vidjetiga misollar

Keling, vidjetdan foydalanishga misollar keltiraylik QLineEdit.

1) Oddiy PyQt QLineEdit misoli

Quyidagi dastur vidjetni qanday yaratishni ko'rsatadi QLineEdit:

```
import sys
from PyQt6.QtWidgets import (
    QApplication,
    QWidget,
    QLineEdit,
    QVBoxLayout
)

class MainWindow(QWidget):
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)

        self.setWindowTitle('PyQt QLineEdit Widget')
        self.setGeometry(100, 100, 320, 210)

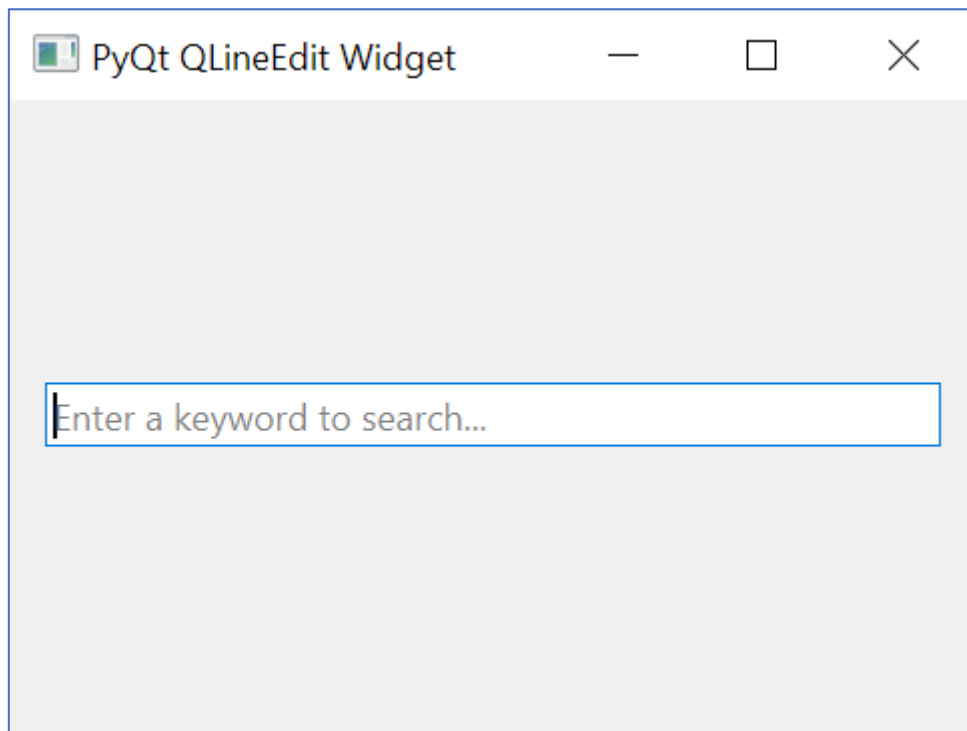
        search_box = QLineEdit(
            self,
            placeholderText='Enter a keyword to search...',
            clearButtonEnabled=True
        )

        # place the widget on the window
        layout = QVBoxLayout()
        layout.addWidget(search_box)
        self.setLayout(layout)

        # show the window
        self.show()

if __name__ == '__main__':
    app = QApplication(sys.argv)
    window = MainWindow()
    sys.exit(app.exec())
```

Chiqish:



2) Parol yozuvini yaratish uchun PyQt QLineEdit dan foydalanish
Quyidagi dastur QLineEditparol kiritish sifatida yangi vidjet yaratadi:

```
import sys
from PyQt6.QtWidgets import (
    QApplication,
    QWidget,
    QLineEdit,
    QVBoxLayout
)

class MainWindow(QWidget):
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)

        self.setWindowTitle('PyQt QLineEdit Widget')
        self.setGeometry(100, 100, 320, 210)

        password = QLineEdit(self, echoMode=QLineEdit.EchoMode.Password)

        # place the widget on the window
        layout = QVBoxLayout()
        layout.addWidget(password)
        self.setLayout(layout)

        # show the window
        self.show()

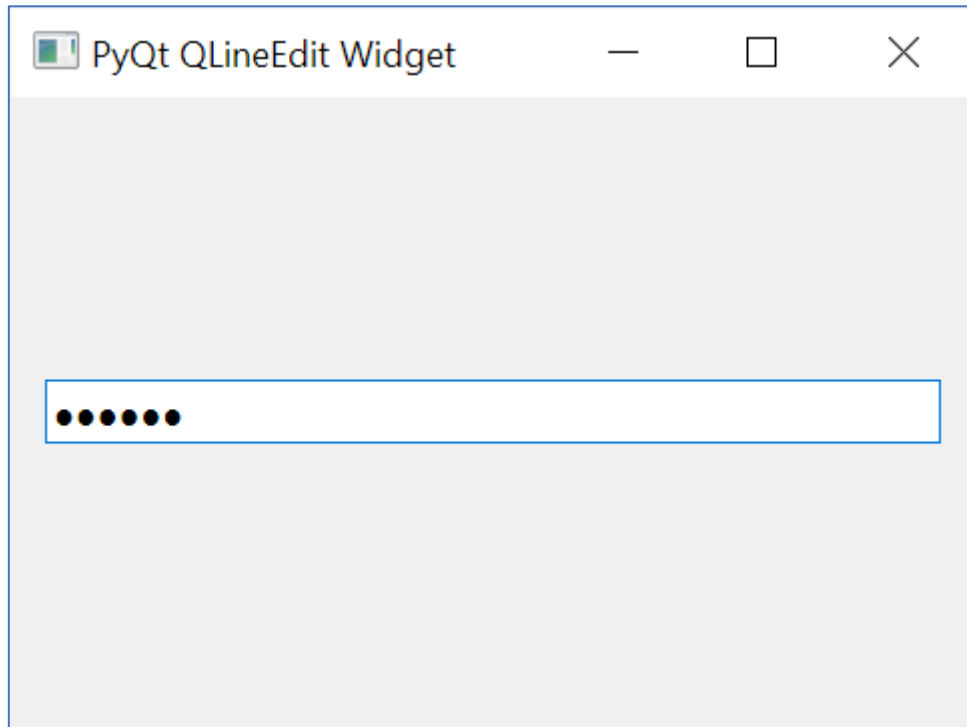
if __name__ == '__main__':
    app = QApplication(sys.argv)
```

```
window = MainWindow()
sys.exit(app.exec())Kod tili: Python ( python )
```

QLineEditVidjetni parol kiritishiga aylantirish uchun siz echoModeni oʻrnatasiz QLineEdit. EchoMode.Password:

```
password = QLineEdit(self, echoMode=QLineEdit.EchoMode.Password)Kod
tili: Python ( python )
```

Chiqish:



3) PyQt QLineEdit-dan avtomatik toʻldirish xususiyati bilan foydalanish

Avtomatik toʻldirish xususiyati bilan yozuv yaratish uchun siz quyidagi amallarni bajaring:

Birinchidan, modulni import QCompleterqiling PyQt6.QtWidgets.

Ikkinchidan, QCompleteravtomatik toʻldirish funksiyasi uchun ishlatiladigan satrlar roʻyxati bilan vidjet yarating:

```
completer = QCompleter(word_list)Kod tili: Python ( python )
```

Uchinchidan, a yarating va toʻldiruvchi obyekt bilan QLineEdituning usulini chaqiring :setCompleter()

```
line_edit = QLineEdit(self)
line_edit.setCompleter(completer)Kod tili: Python ( python )
```

Masalan, quyidagi dastur QLineEditavtomatik toʻldirish xususiyatiga ega vidjetni koʻrsatadi:

```
import sys
from PyQt6.QtWidgets import (
    QApplication,
    QWidget,
    QLineEdit,
    QVBoxLayout,
```

```

        QCompleter

    )

    class MainWindow(QWidget):
        def __init__(self, *args, **kwargs):
            super().__init__(*args, **kwargs)

            self.setWindowTitle('PyQt QLineEdit Widget')
            self.setGeometry(100, 100, 320, 210)

            common_fruits = QCompleter([
                'Apple',
                'Apricot',
                'Banana',
                'Carambola',
                'Olive',
                'Oranges',
                'Papaya',
                'Peach',
                'Pineapple',
                'Pomegranate',
                'Rambutan',
                'Ramphal',
                'Raspberries',
                'Rose apple',
                'Starfruit',
                'Strawberries',
                'Water apple',
            ])
            fruit = QLineEdit(self)
            fruit.setCompleter(common_fruits)

            # place the widget on the window
            layout = QVBoxLayout()
            layout.addWidget(fruit)
            self.setLayout(layout)

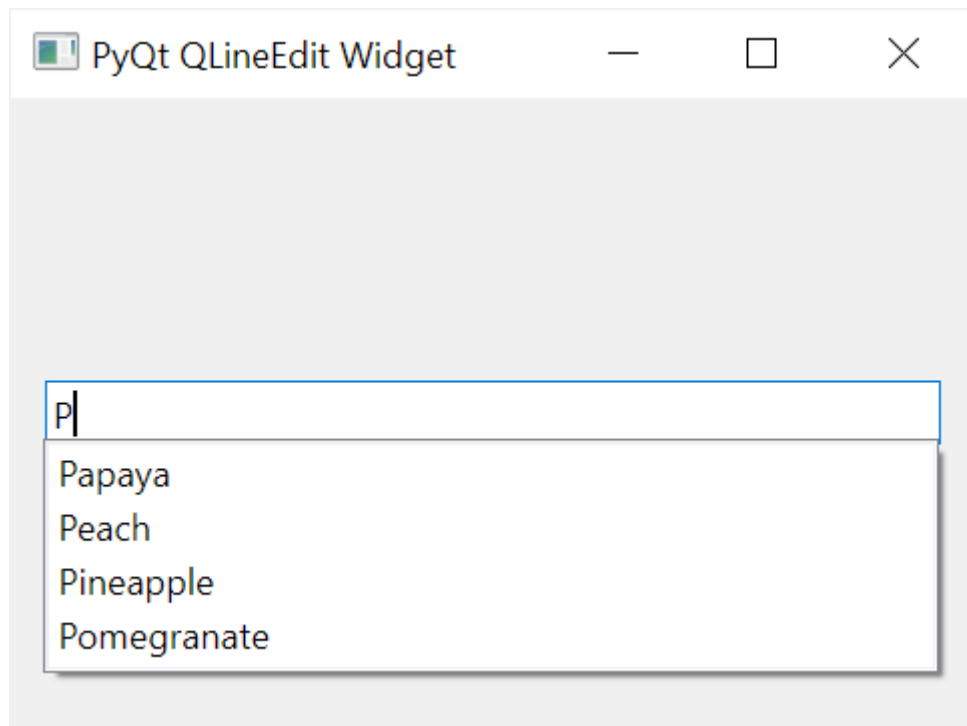
            # show the window
            self.show()

if __name__ == '__main__':
    app = QApplication(sys.argv)
    window = MainWindow()
    sys.exit(app.exec())

```

Chiqish:

Kod tili: Python (python)



- QLineEditBir qatorli kirish vidjetini yaratish uchun foydalaning .
- echoModeMatnni ko'rsatish usulini o'zgartirish uchun xususiyatdan foydalaning .
- QLineEditAvtomatik to'ldirish funksiyasini qo'llab-quvvatlash uchun vidjetdan QCompleter vidjeti bilan foydalaning .

Nazorat savollari:

1. PyQt da yorliq yaratish uchun qaysi vidgetdan foydalaniladi?
2. QLineEdit vidjetining vazifasi qanday?
3. QLineEdit vidjetining xususiyatlari qanday?